

**Savitribai Phule Pune University**  
**T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination**  
**Lab Course CS-305-MJ-P**  
**Operating System-I**

**Duration: 3 Hours****Maximum Marks:35**

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Q.1 Write a program to simulate Linked file allocation method. Assume disk with 'n' number of blocks. Give value of 'n' as input. Randomly mark some block as allocated and accordingly maintain the list of free blocks Write menu driven program with menu options as mentioned below and implement each option.

- a. Show Bit Vector
- b. Create New File
- c. Show Directory
- d. Delete File
- e. Exit

[15 Marks]

Q.2 Write the program to simulate Non-preemptive Priority scheduling.

The arrival time and first CPU-burst for different n number of processes should be input to the algorithm. The next CPU-burst should be generated randomly. The output should give Gantt chart, turnaround time and waiting time for each process. Also find the average waiting time and turnaround time.

[15 Marks]

Q.3. Oral/Viva

[05 Marks]